



OBJECTIVE:

Predict whether an eCommerce customer session will result in a future purchase after observing only the first two session events.

FINAL MODEL:

LightGBM

PREDICATION TARGET:

future_purchase

Problem Formulation

Business Question

Can early customer-session behavior predict whether a user will purchase later in the same session?

ML Framing

Component	Definition
Task	Binary classification
Target	future_purchase
Positive class	Purchase after first two events
Prediction point	After event 2
Main metric	PR-AUC / Average Precision

Why PR-AUC?

Future purchase is a minority outcome, so accuracy alone is not enough.

Data Funnel

Dataset Construction

Stage	Count
Raw event records scanned	66,589,268
Session-preserved sampled events	982,585
Unique sampled sessions	173,519
Leakage-safe modeling sessions	92,799
Final future-purchase rate	12.58%

Key Design Choice

A session-preserved sample was used so each selected session kept its full event sequence.

Leakage-Safe Target Design

Observation Window

Only the first two events were used as predictors.

Target Window

The model predicts whether a purchase occurs after the second event.

Rule	Purpose
Use first two events only	Prevent target leakage
Exclude purchases inside observation window	Avoid predicting already-known purchases
Chronological train/validation/test split	Simulate future prediction

Timeline

Event 1 + Event 2 → Prediction → Later Events → Future Purchase Target

Cleaning and Preprocessing

Cleaning Decisions

Issue	Treatment	Justification
Exact duplicate rows	Removed 1,921 rows	Avoid inflated behavior counts
Missing brand/category	Standardized as unknown	Missing values may carry signal
Zero-price events	Retained	Valid customer interactions
Price aggregation	Positive prices only	Prevent zero prices distorting price features
High-price outliers	Retained	May represent valid catalog items

Validation

Deduplication did not change any session-level conversion labels.

EDA: Early Behavior Signals

Key Findings

- Complete-session conversion rate: **6.76%**
- Leakage-safe future-purchase rate: **12.58%**
- Early cart activity had the strongest behavioral relationship with future purchase
- Faster early-session movement was associated with higher future-purchase likelihood

Technical Implication

Early behavior is predictive enough to support a two-event observation window.

Feature Engineering

Feature Families

Feature Family	Examples
Early behavior	early_cart_count, early_unique_products
Product diversity	early_unique_categories, early_unique_brands
Product context	first/second category and brand
Price	first/second positive observed prices
Timing	early_event_gap_log1p, first event hour
Calendar	day of week, weekend flag

Leakage Control

No features were derived from events after the second event.

PCA and Feature Selection

PCA

- Fitted on training data only
- Preprocessed feature count: **511**
- PC1 + PC2 explained variance: **29.79%**
- 26 components needed for 80% variance
- PCA used for exploration, not final modeling

Embedded Feature Selection

- L1 Logistic Regression used on training data only
- Raw candidate features: **17**
- Retained features: **16**
- Excluded feature: event_type_first

Conclusion

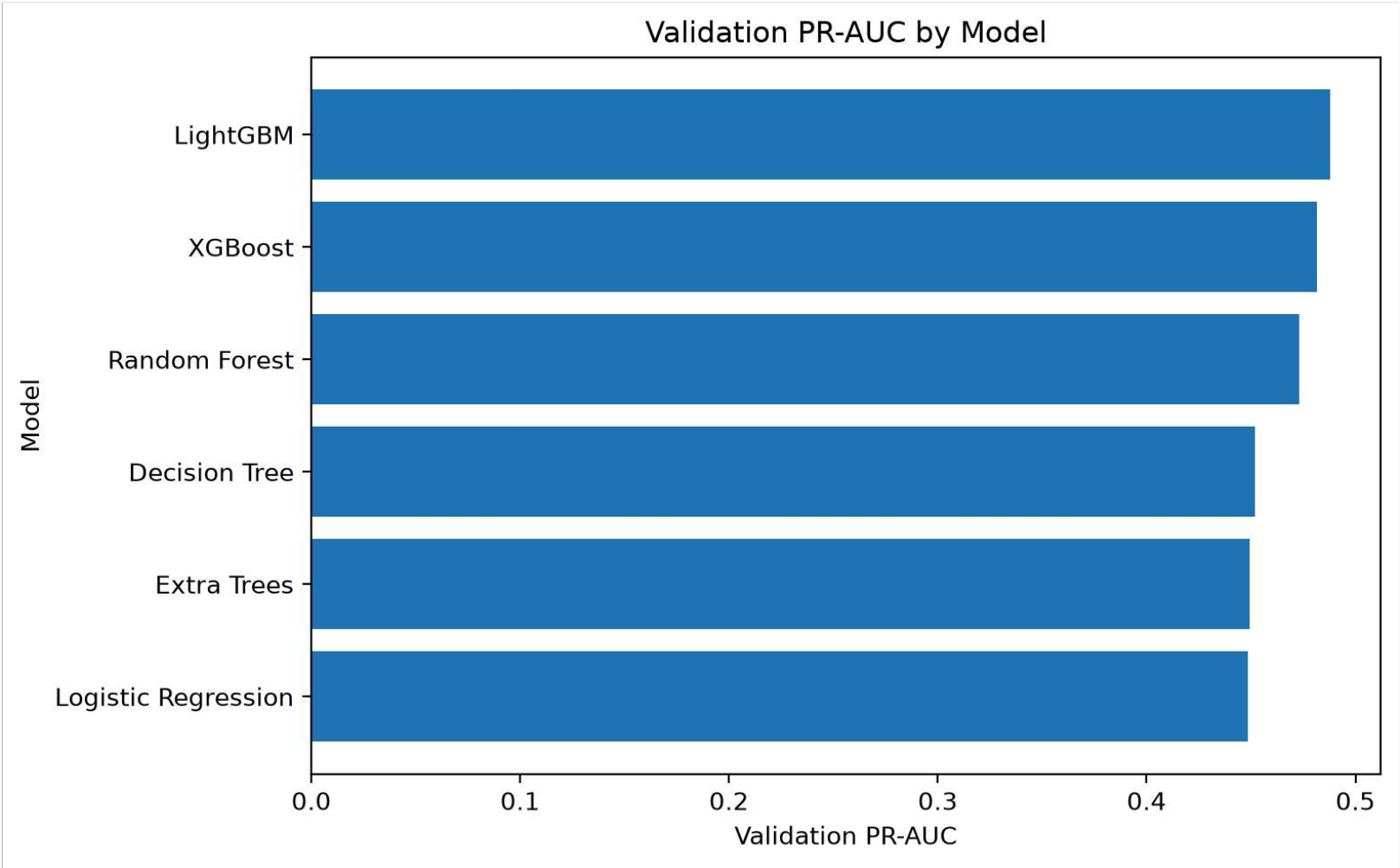
Original engineered features were more interpretable.

Model Comparison

Models Evaluated

- Logistic Regression
- Decision Tree
- Random Forest
- Extra Trees
- XGBoost
- LightGBM

Validation PR-AUC



Selection

LightGBM achieved the strongest validation PR-AUC.

Final Model Evaluation

Final Model

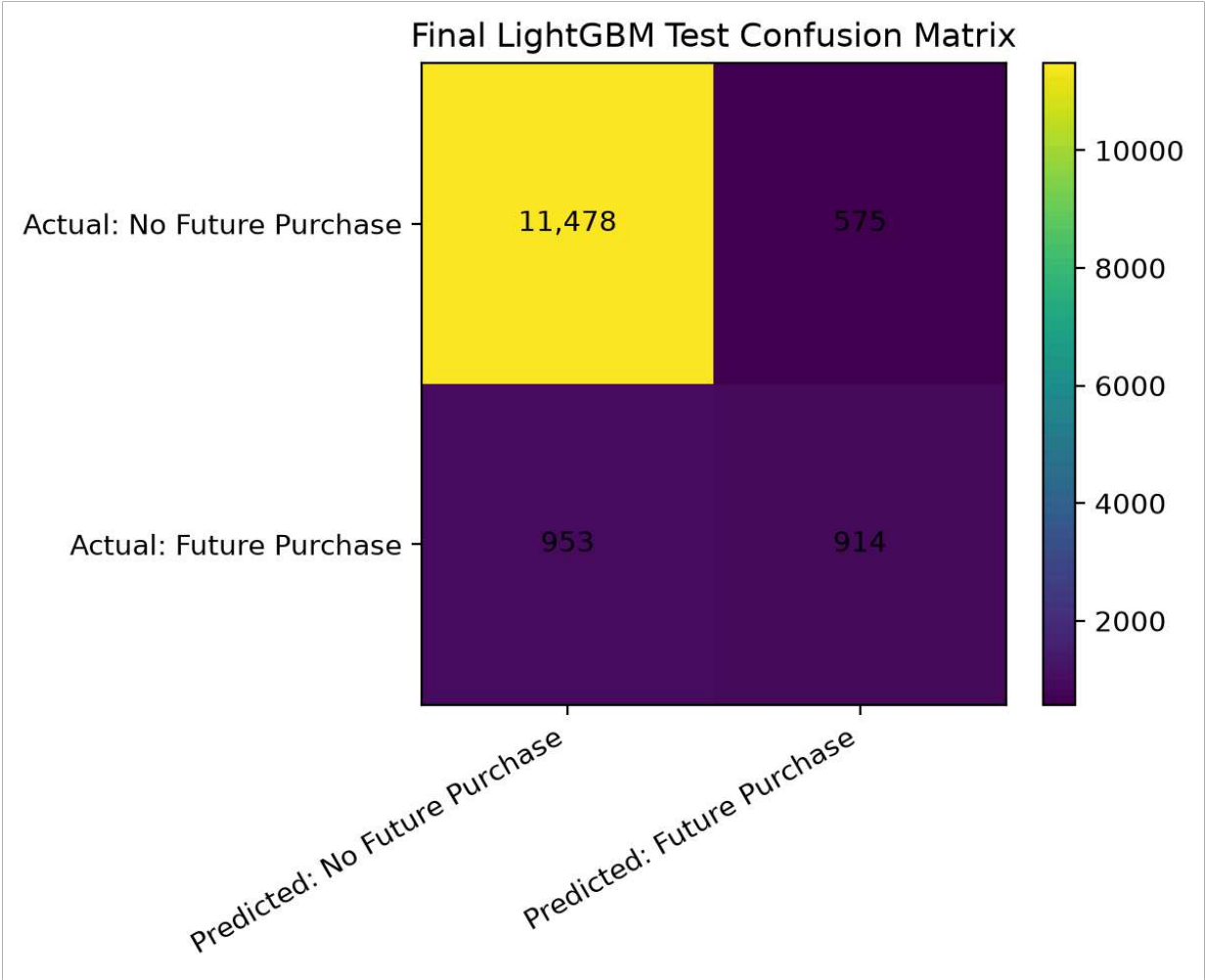
LightGBM, threshold = **0.65**

Metric	Test Result
PR-AUC	0.4824
ROC-AUC	0.7802
Accuracy	0.8902
Precision	61.38%
Recall	48.96%
F1-score	0.5447
Sessions targeted	1,489
Targeting rate	10.70%

Interpretation

The model identifies a smaller high-propensity group with strong precision.

Final Test Confusion Matrix



Interpretation

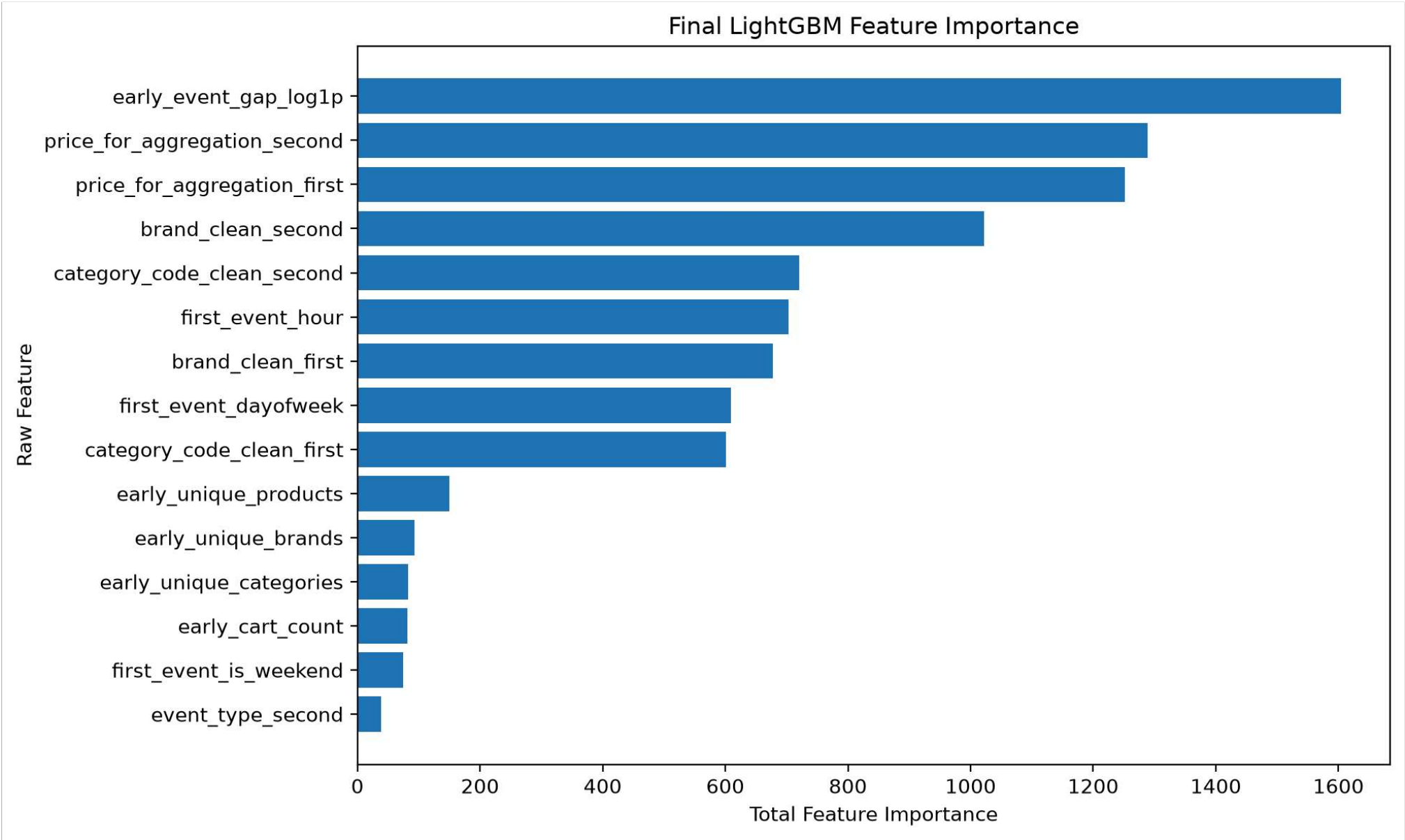
- True positives: **914**
- False positives: **575**
- False negatives: **953**
- True negatives: **11,478**

Technical Trade-off

The threshold prioritizes precision and targeted-session efficiency, but still misses some future purchasers.

Feature Importance

Final LightGBM Feature Importance



Top Predictors

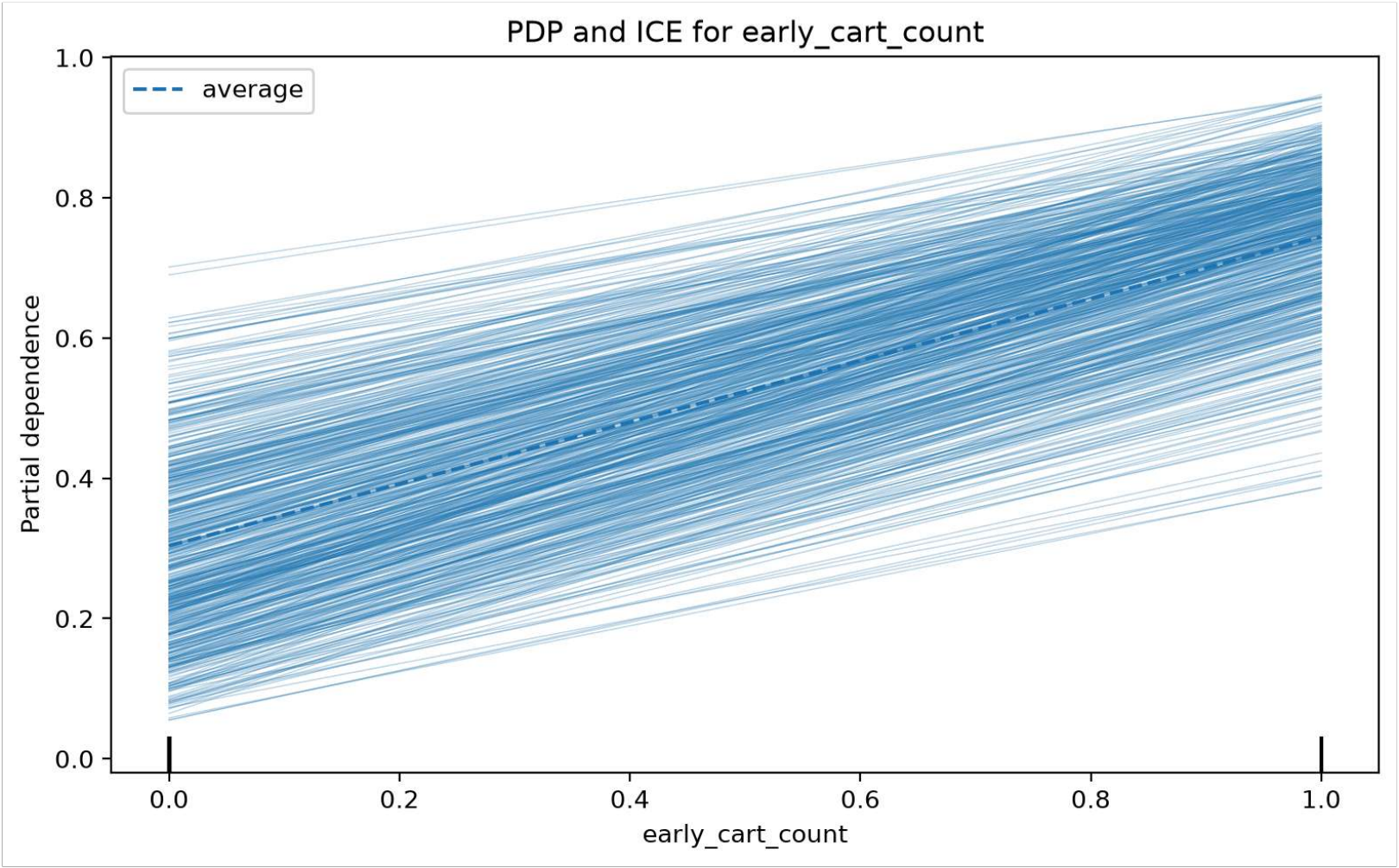
- 1. early_event_gap_log1p
- 2. price_for_aggregation_second
- 3. price_for_aggregation_first
- 4. brand_clean_second
- 5. category_code_clean_second

Interpretation

The model relies on early session speed, price exposure, brand, category, and timing context.

PDP / ICE Explainability

Early Cart Count

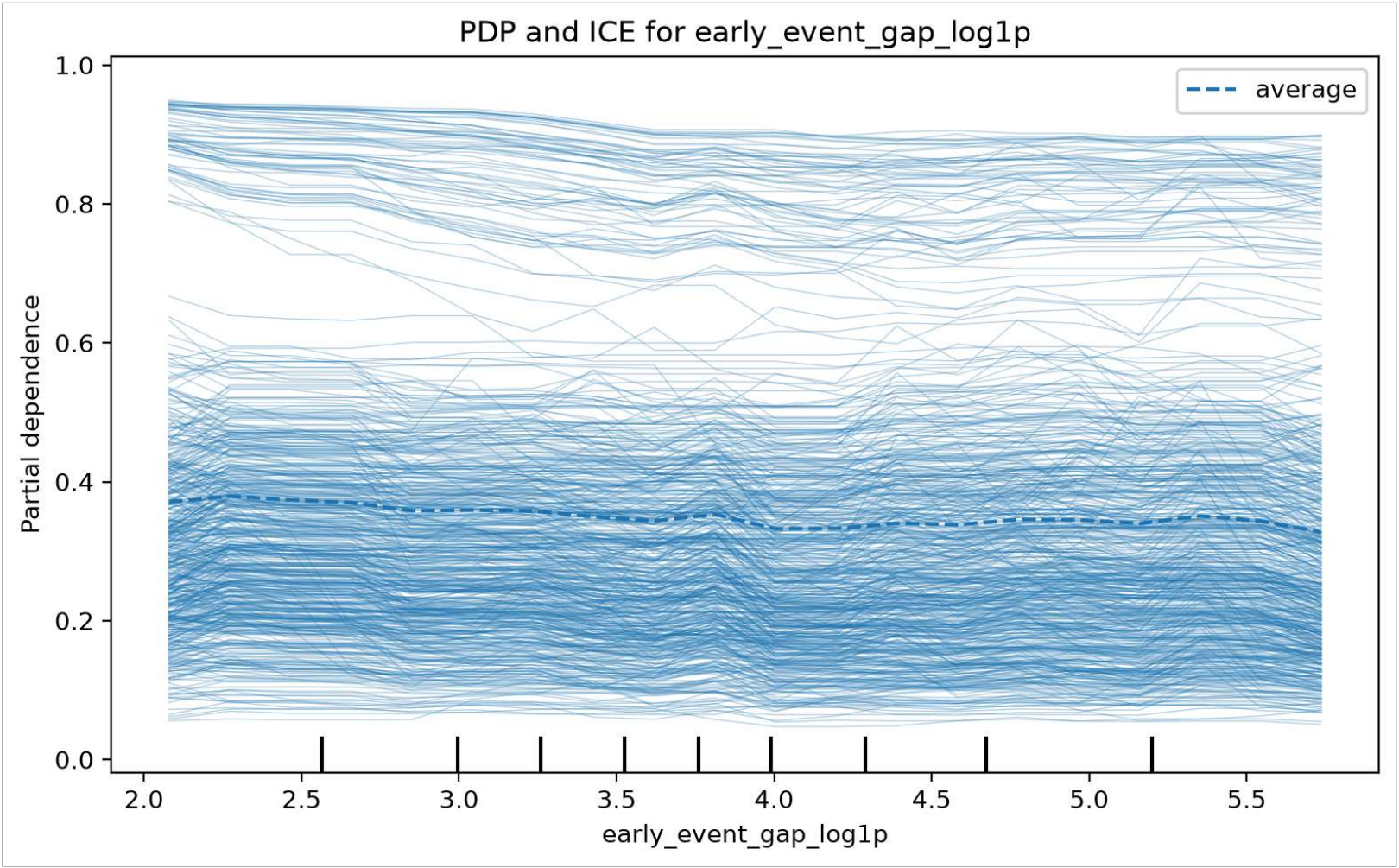


Interpretation

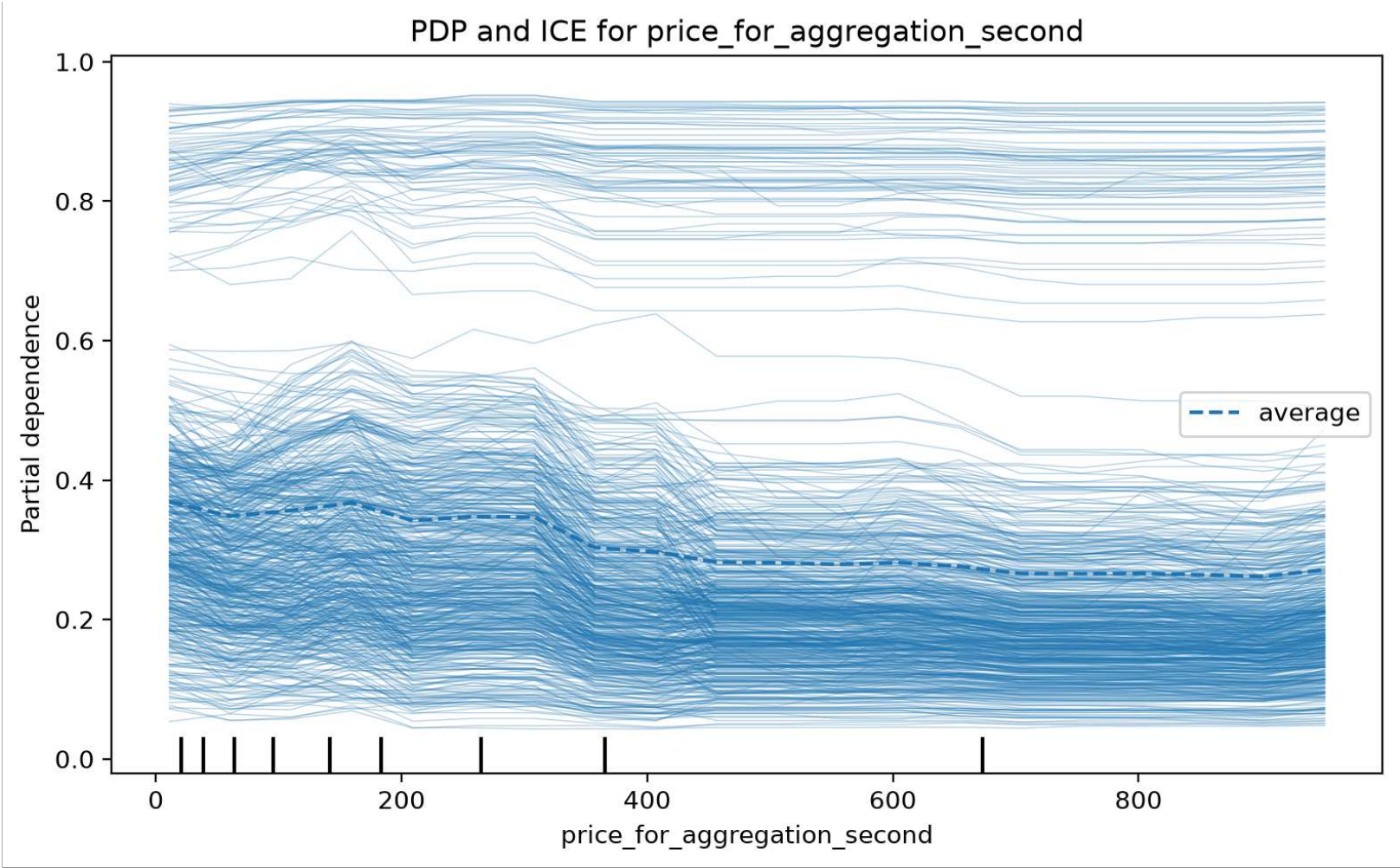
The model sharply increases predicted future-purchase probability when a cart event appears within the first two events. This confirms the EDA and segment-audit finding that early cart activity is a strong purchase-intent signal.

PDP / ICE: Event Gap and Price

Event Gap



Second Observed Price



Interpretation

- Longer gaps between the first two events generally reduce predicted purchase probability.
- Higher second observed price tends to lower predicted probability in the model.
- PDP/ICE explains model behavior, not causality.

Ethical AI and Bias Audit

Sensitive Attribute Review

The dataset does not contain:

- gender
- race
- age
- income
- socioeconomic status

Fairness Limitation

Formal demographic fairness metrics could not be calculated.

Segment-Level Audit

Instead, the model was audited across:

- product category
- brand
- price band
- first-event timing
- early cart behavior
- early event-gap behavior

Key Risk

The model favors immediate-intent sessions and may under-target slower exploratory shoppers.

Technical Limitations

Main Limitations

- Dataset covers only April 2020
- No direct demographic fairness audit possible
- Class imbalance remains important
- False negatives are still significant
- Model behavior may drift over time
- Feature importance and PDP/ICE are not causal explanations

Responsible Use

The model should be used for prioritization, not exclusion.

Conclusion and Future Work

Final Conclusion

A leakage-safe LightGBM model was developed to predict future purchase intent after only two session events.

Technical Outcome

- Best validation model: **LightGBM**
- Final test PR-AUC: **0.4824**
- Final test precision: **61.38%**
- Targeted only **10.70%** of test sessions

Future Work

- Validate on newer months of data
- Add customer-level history if available
- Monitor performance by segment
- Recalibrate thresholds by campaign objective
- Retrain periodically to handle drift